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Student information

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Presentation goal

Study whether a generalizable multi-view 3D reconstruction model can recover scene geometry from only a few overlapping images of a new scene, without per-scene optimization at test time.

- 3D reconstruction is important because 3D representations help machines understand spatial structure, depth, and object relationships more accurately.
- Practical 3D capture often has only a small number of usable images.
- Classical high-quality pipelines depend on accurate camera poses or expensive scene-specific optimization.
- Sparse views increase ambiguity because cross-view correspondence is harder to establish reliably.
- A fast, generalizable method for unseen scenes would be useful for robotics, AR/VR, navigation, and scalable environment digitization.

Core question

How can we reconstruct unseen scenes accurately and efficiently when only 2 to 5 input views are available?

Key constraints

- Unseen scene at test time
- Sparse overlapping images
- No per-scene optimization

Expected outcome

- Robust geometry recovery
- Stable performance under view reduction
- Practical runtime per scene

1. Evaluate a generalizable 3D reconstruction model on unseen scenes.
2. Analyze sparse-view performance with 2, 3, and 5 images.
3. Compare a pairwise baseline (DUSt3R) with a single-stage multi-view method (MV-DUSt3R+).
4. Identify conditions where sparse-view reconstruction fails or remains reliable.

Working hypothesis

A feed-forward multi-view model should be more robust than a pairwise baseline when the number of input views is limited, because it reasons over multiple views jointly in one pass.

- Better global consistency across views
- Less dependence on separate alignment stages
- Stronger behavior in highly ambiguous sparse-view settings

Our contribution

Beyond benchmarking DUSt3R and MV-DUSt3R+, we first identify which sparse-view components help, then propose a supervised extension for the failure cases that heuristic filters could not solve.

- **View selection for far-reference issue:** replace random sparse-view sampling with policies that balance scene coverage, viewpoint diversity, and overlap.
- **Honest ablation of heuristic fusion/filtering:** test confidence, multi-view support, occlusion, and ambiguity signals separately.
- **Supervised reliability extension:** replace failed hand-written filters with learned occlusion reliability and repeated-structure match disambiguation heads.

1. View selection for far-reference issue

- Compare **random baseline**, **coverage-aware**, **diversity-aware**, and **hybrid** sparse-view policies.
- The goal is to reduce far-reference failures while preserving co-visibility under 2–5 input views.

2. Heuristic ablation before adding learned modules

- Evaluate C, S_{mv}, M, occlusion penalties, and ambiguity penalties independently.
- If a module reduces F-score or completeness, keep it out of the final pipeline and report the failure explicitly.

3. Supervised occlusion and ambiguity heads

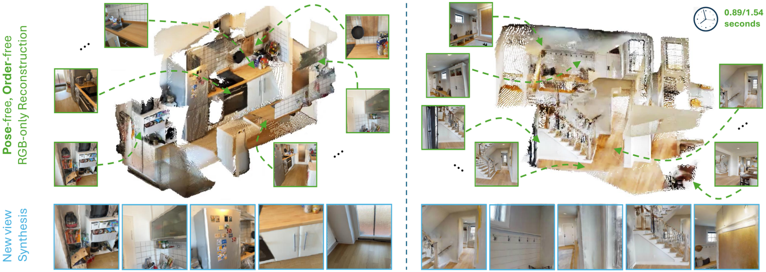
- Train an **Occlusion-Aware Reliability Head** to keep occluded-but-valid points and reject floating/wrong geometry.
- Train a **Repeated-Structure Disambiguation Head** using reciprocal matches, descriptor margin, reprojection, and cycle error.

Why not keep tuning heuristic filters?

Early ablations show that hand-written occlusion and ambiguity filters can remove valid geometry and reduce completeness. The next step is supervised reliability learning rather than stronger manual penalties.

- **OARH:** predict whether each candidate point should be kept, using confidence, visibility, occlusion, reprojection, depth disagreement, and multi-view agreement features.
- **RSDH:** predict whether repeated-structure matches are valid, using MASt3R reciprocal matching, feature margin, 3D disagreement, and cycle consistency.
- **Fine-tuning policy:** freeze MV-DUSt3R+ first; only later unfreeze the confidence head and last decoder blocks if validation improves.

- Primary method: **MV-DUSt3R+**
- Single-stage feed-forward multi-view reconstruction
- Targeted at sparse-view scenes without camera calibration
- Our study focuses on robustness, quality, and runtime



Source: Tang et al., MV-DUSt3R+, project page and paper

DUSt3R baseline

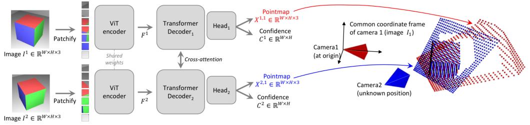
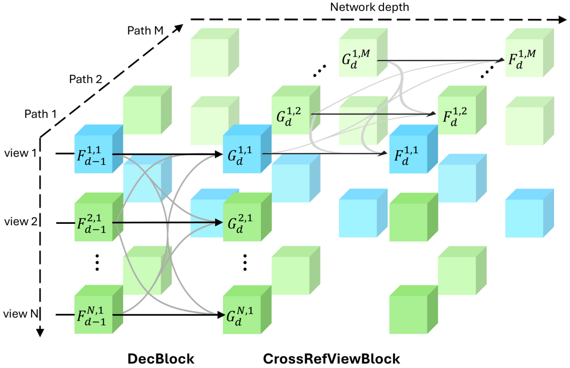


Figure 2. **Architecture of the network.** Two views of a scene (I^1, I^2) are first encoded in a Siamese manner with a shared ViT encoder. The resulting token representations F^1 and F^2 are then passed to two transformer decoders that constantly exchange information via cross-attention. Finally, two regression heads output the two corresponding pointmaps and associated confidence maps. Importantly, the two pointmaps are expressed in the same coordinate frame of the first image I^1. The network is trained using a simple regression loss (Eq. (4))

Pairwise reasoning with cross-attention between two views

MV-DUSt3R+ main method



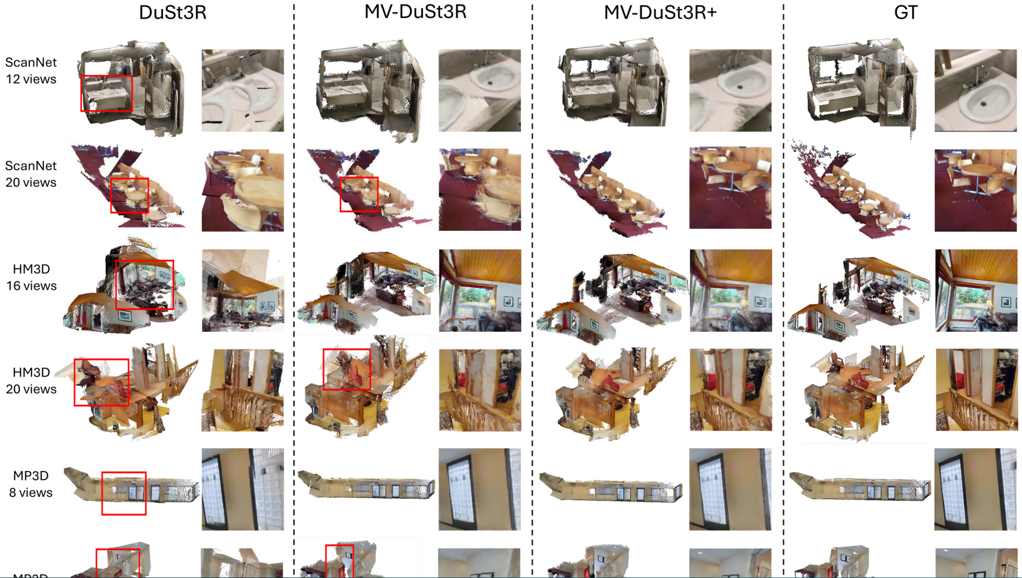
Multi-path design with cross-reference-view information flow

Source: Left: Wang et al., DUSt3R, CVPR 2024. Right: Tang et al., MV-DUSt3R+, project page/paper

Aspect	DUSt3R	MV-DUSt3R+
Processing style	Pairwise reconstruction	Joint multi-view reconstruction
Global consistency	Requires later alignment	Built into one forward pass
Sparse-view behavior	Can accumulate pairwise errors	More robust cue fusion
Runtime trend	Slower as views increase	Better scaling with more views
Role in project	Reference baseline	Main candidate method

- DUSt3R is still a strong baseline, so the comparison is direct and meaningful.
- MV-DUSt3R+ is better aligned with our target setting of 2–5 sparse views.

	Method	GO	HM3D			ScanNet			MP3D			Time
			ND ↓	DAc ↑	CD ↓	ND ↓	DAc ↑	CD ↓	ND ↓	DAc ↑	CD ↓	(sec)
4 views	Spann3R	×	37.1	0.0	225(184)	8.9	19.5	54.7(50.1)	42.7	0.0	248(202)	0.36
	DUSt3R	✓	1.9	75.1	5.6(2.3)	1.3	89.8	4.0(0.4)	3.9	41.7	40.0(5.3)	2.42
	MV-DUSt3R	×	1.1	92.2	2.0(1.1)	1.0	93.3	2.0(0.4)	2.5	62.4	25.3(4.1)	0.05
	MV-DUSt3R+	×	1.0	95.2	1.5(0.9)	0.8	94.9	1.5(0.3)	2.2	68.0	19.9(3.4)	0.29
	MV-DUSt3R$_{\text{oracle}}$	×	1.0	94.6	1.5(0.7)	0.8	95.5	1.3(0.3)	2.3	66.6	20.7(4.0)	-
	MV-DUSt3R+$_{\text{oracle}}$	×	0.9	96.5	1.4(0.7)	0.7	95.8	1.2(0.2)	2.1	70.6	17.9(3.3)	-
12 views	Spann3R	×	32.6	0.0	125(113)	9.1	16.3	36.6(31.2)	35.0	0.0	138(112)	1.34
	DUSt3R	✓	3.9	30.7	18.1(3.4)	1.9	82.6	4.1(0.6)	6.6	12.0	49.6(8.3)	8.28
	MV-DUSt3R	×	1.6	79.5	3.0(1.2)	1.4	86.8	2.3(0.8)	3.4	41.3	22.6(5.5)	0.15
	MV-DUSt3R+	×	1.2	91.5	1.8(0.7)	1.2	88.4	1.8(0.7)	2.6	55.0	15.1(3.8)	0.89
	MV-DUSt3R$_{\text{oracle}}$	×	1.3	88.8	1.8(0.9)	1.0	90.6	1.3(0.7)	2.9	51.3	16.4(4.0)	-
	MV-DUSt3R+$_{\text{oracle}}$	×	1.1	94.8	1.4(0.7)	1.0	90.9	1.3(0.5)	2.5	59.8	13.6(3.5)	-
24 views	Spann3R	×	41.7	0.0	139(121)	11.4	1.6	37.4(35.5)	46.6	0.0	151(121)	2.73
	DUSt3R	✓	6.8	7.3	32.4(5.2)	2.4	72.6	5.1(1.0)	11.4	2.5	80.9(14.3)	27.21
	MV-DUSt3R	×	3.4	36.7	10.0(3.5)	2.2	75.2	2.7(0.9)	6.3	12.2	38.6(13.9)	0.35
	MV-DUSt3R+	×	2.1	64.5	3.9(2.0)	1.6	81.2	1.7(0.7)	4.3	26.7	22.0(5.9)	1.97
	MV-DUSt3R$_{\text{oracle}}$	×	2.1	58.9	3.5(2.1)	1.4	82.9	1.4(0.7)	4.4	22.0	19.9(5.1)	-
	MV-DUSt3R+$_{\text{oracle}}$	×	1.8	77.9	2.6(1.3)	1.3	85.1	1.3(0.6)	3.6	33.1	15.1(4.4)	-


DuSt3R
MV-DuSt3R
MV-DuSt3R+
GT
ScanNet
12 views
ScanNet
20 views
HM3D
16 views
HM3D
20 views
MP3D
8 views

Prepare multi-view scenes and split into train / validation / unseen test

Create sparse-view subsets with 2, 3, 4, and 5 input images

Run MV-DUSt3R+ pretrained with selected sparse views

Apply learned reliability / visibility / ambiguity heads

Evaluate geometry, robustness, runtime, and case-specific failures

Controlled factors

- Number of input views
- Scene-level train / validation / test split
- Fixed threshold on final test

Observed outcomes

- Reconstruction accuracy
- Completeness and consistency
- Runtime per scene
- Occlusion and repeated-structure failures

1. First build a supervised label cache with per-view visibility, occlusion, floating/wrong-depth, and match labels.
2. Mine occlusion-heavy, low-overlap, and hard-negative subsets before training the learned heads.
3. Freeze MV-DUSt3R+ and train small reliability / match heads on generated GT labels.
4. Evaluate OARH on occlusion-heavy scenes and RSDH on repeated-structure scenes before mixing them into the full pipeline.
5. If validation improves, unfreeze only the confidence head and last decoder blocks for light fine-tuning.
6. Keep final test clean: no threshold tuning, no checkpoint selection on test scenes.

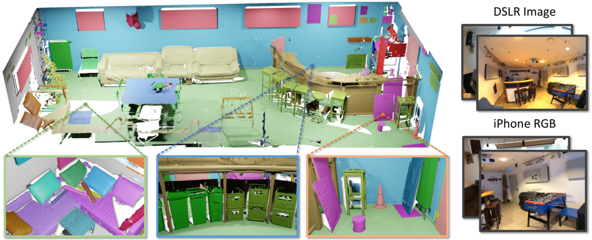
Primary dataset: ScanNet++

High-fidelity indoor scenes with registered DSLR images and commodity iPhone captures.

Secondary dataset: DTU

Controlled object-centric scenes for supplementary evaluation and sanity checks in a cleaner setting.

If compute is limited, we will use curated subsets rather than the full datasets.



Source: Yeshwanth et al., ScanNet++, ICCV 2023, Figure 1

Dataset	Num. scenes	Average scans/scene	Total scans	RGB (MP) Commodity	RGB (MP) DSLR	Depth LR	Depth HR	Dense semantics	NVS
LLFF [26]	35	-	-	12	✗	✗	✗	✗	✓
DTU [15]	124	-	-	1.9	✗	✗	✓	✗	✓
BlendedMVS [47]	113	-	-	✗	✗	✗	✗	✓	✗
ScanNet [9]	1503	-	-	1.25	✗	✓	✗	✓	✗
Matterport3D [4]	90‡	-	-	1.3	✗	✓	✗	✓	✗
Tanks and Temples [20]	21	10.57	74	✗	8	✗	✓	✗	✓
ETH3D [35]	25	2.33	42	✗	24	✗	✓	✗	✗
ARKitScenes [3]	1004	3.16	3179	3	✗	✓	✓	✗	✗
ScanNet++ (ours)	460	4.85	1858	2.7	33	✓	✓	✓	✓

Source: Yeshwanth et al., ScanNet++, ICCV 2023, Table 1

- 460 indoor scenes with high-fidelity geometry
- DSLR images support stronger geometric supervision
- Commodity captures make sparse-view testing realistic
- Dense semantics and NVS setup make the benchmark richer

- Reconstruction accuracy against available ground-truth geometry
- Completeness and consistency of reconstructed point clouds
- Performance degradation under 2-view, 3-view, and 5-view settings
- Inference efficiency, including runtime per scene

Success criterion

Show that the multi-view method is more robust than the pairwise baseline under identical sparse-view conditions.

Source: Tang et al., MV-DUSt3R+, quantitative benchmark

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Table 2 MVS reconstruction results. Best results are marked in red. For clarity, metric ND is scaled up by 10×, DAc is in percent %, and CD is scaled by 100× with its median given in parentheses. Results of our **oracle** methods are obtained by manually selecting the best single reference view from reference view candidates to report a possibly achievable

- End-to-end implementation for training and evaluation
- Quantitative results on unseen scenes across multiple view counts
- Qualitative 3D visualizations and failure-case analysis
- Final report discussing strengths, limitations, and recommendations
- Reproducible Kaggle experiment scripts organized under `scripts/kaggle`
- Final validation rerun script with a P100-compatible Torch fallback for unstable Kaggle GPU allocation
- Submitted supervised-extension kernels for Run 12–18, including reliability, match-disambiguation, reciprocal-feature, fine-tune-gate, and learned-summary runs
- Submitted Phase 3 label-cache kernel for Run 19 to support stricter occlusion and ambiguity supervision
- Submitted Run 20 subset-mining kernel to prepare balanced OARH v2 and RSDH v2 training/evaluation manifests
- Clean GitHub repository with curated docs, proposal sources, and PDF deck

- A focused benchmark of sparse-view reconstruction on unseen scenes
- A direct comparison between pairwise and single-stage multi-view reasoning
- Evidence about when sparse-view reconstruction remains reliable
- Practical guidance for follow-up work on robust 3D reconstruction
- An ablation record showing which proposed modules help and which fail
- A supervised follow-up plan with honest outcomes: gated OARH still needs better labels/features, while RSDH is strong on a GT-depth-assisted MASt3R proxy but not yet a final image-only ambiguity solution
- A stricter Phase 3 protocol that requires held-out occlusion-heavy and repeated-structure gains before claiming the remaining limitations are solved

- Wang et al., "DUSt3R: Geometric 3D Vision Made Easy," CVPR 2024.
- Tang et al., "MV-DUSt3R+: Single-Stage Scene Reconstruction from Sparse Views In 2 Seconds," CVPR 2025.
- Yeshwanth et al., "ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes," ICCV 2023.
- DTU Robot Image Data Sets, Multi-View Stereo Benchmark.